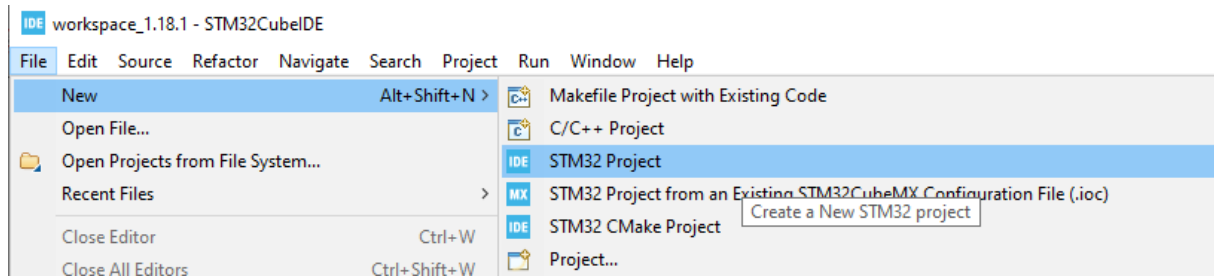


1. Open STM32CubeIDE
2. File->New->STM32 Project

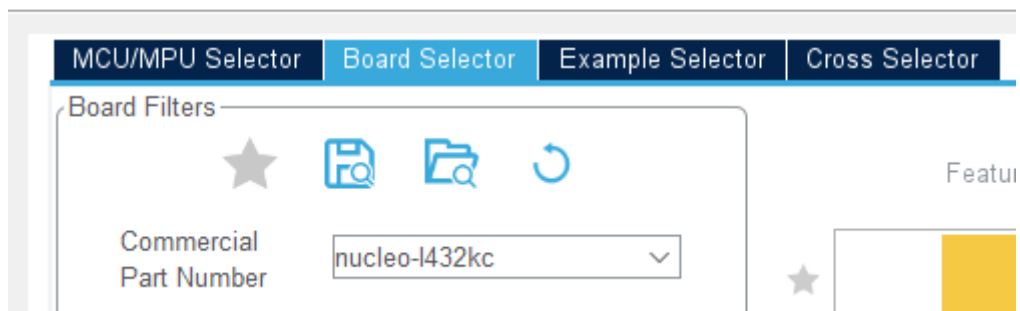


3. Type in Nucleo-L432KC (our board)

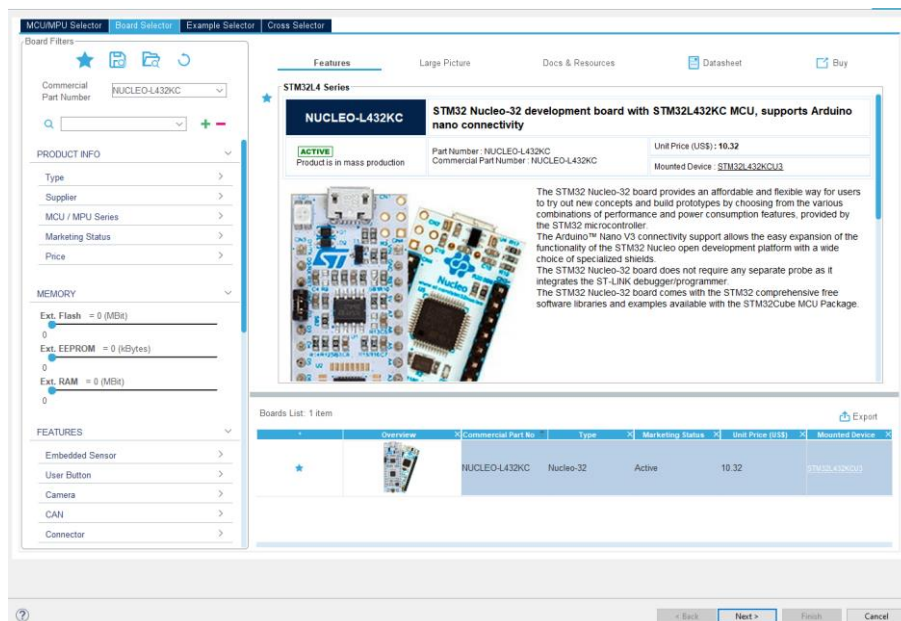
STM32 Project

Target Selection

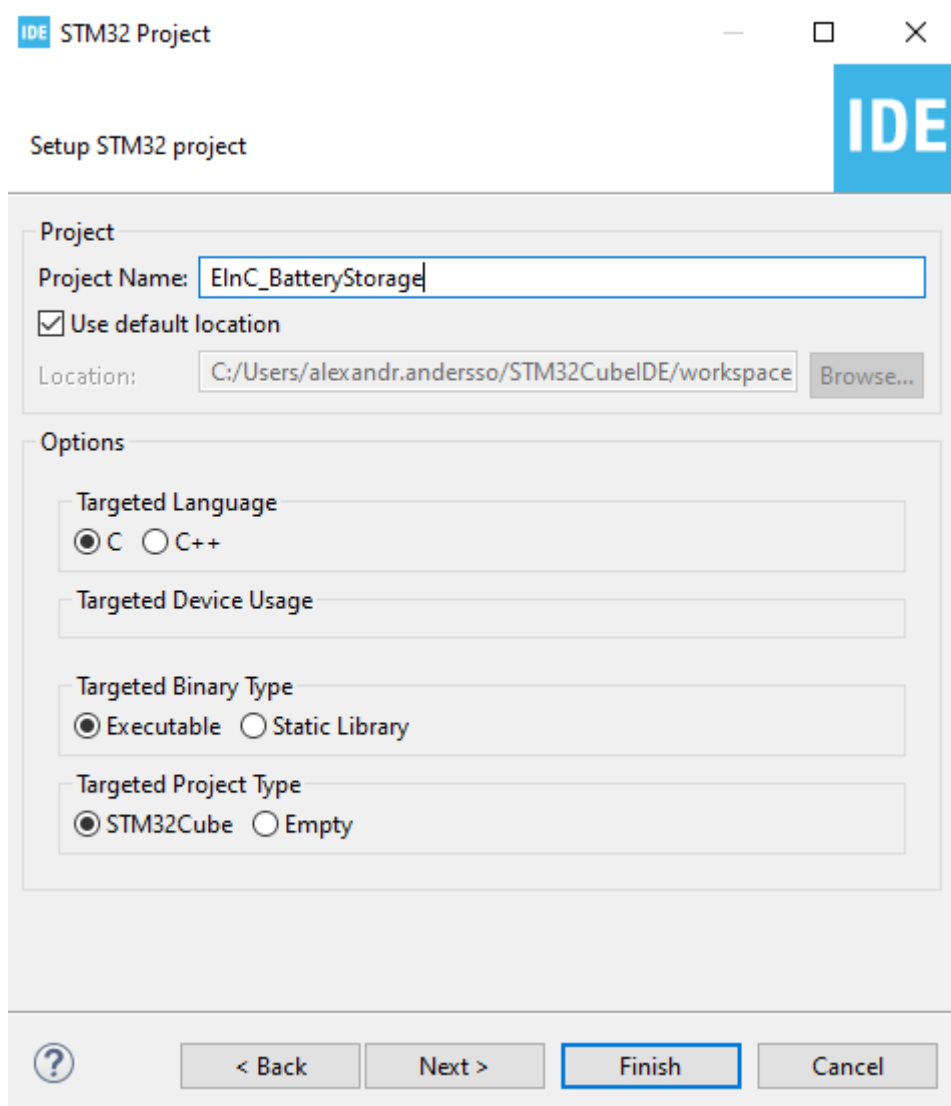
⚠ STM32 target or STM32Cube example selection is required



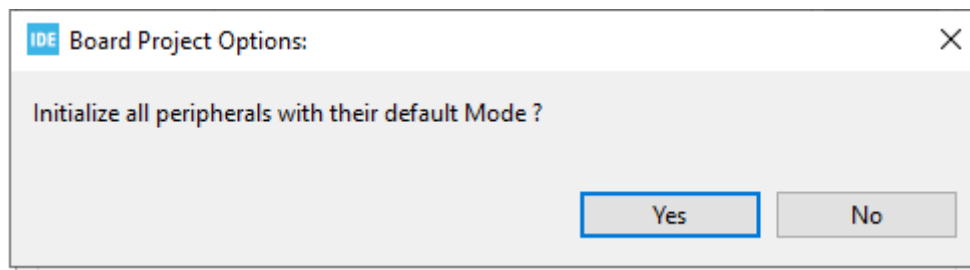
4. Select the board and click Next



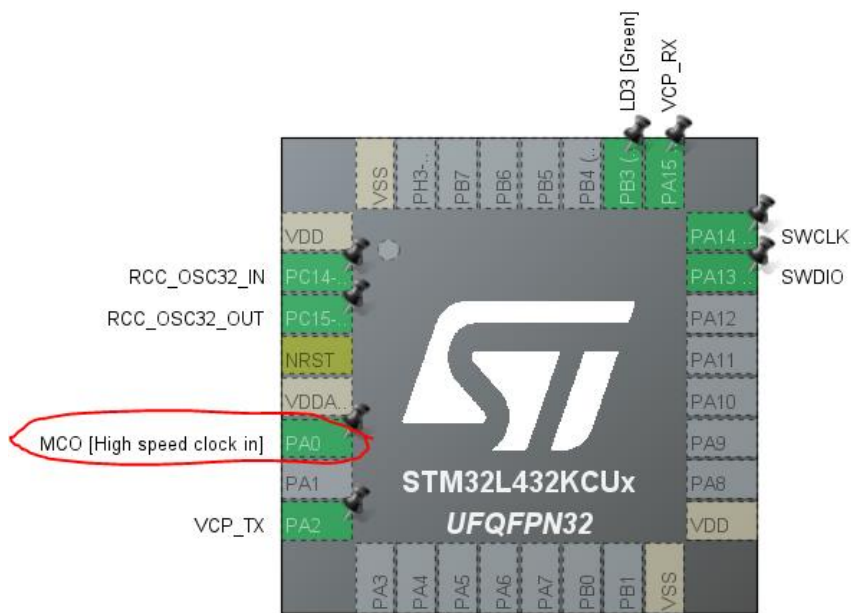
5. Name your project
6. Choose your language (C or C++) as you wish



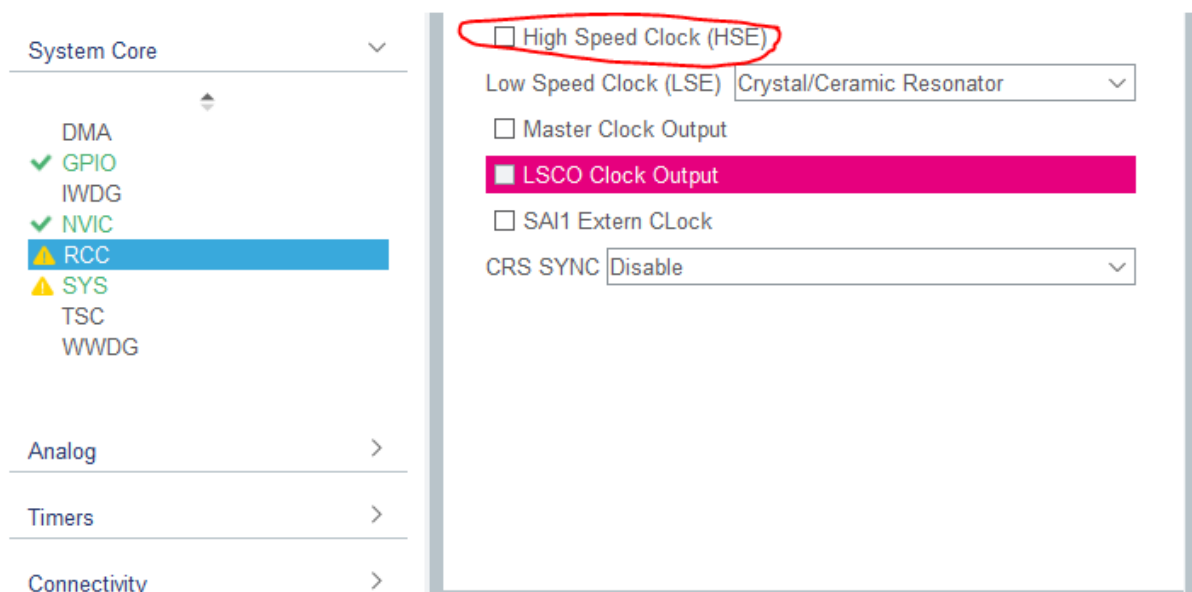
7. Click Yes on Initialize all peripherals with their default Mode



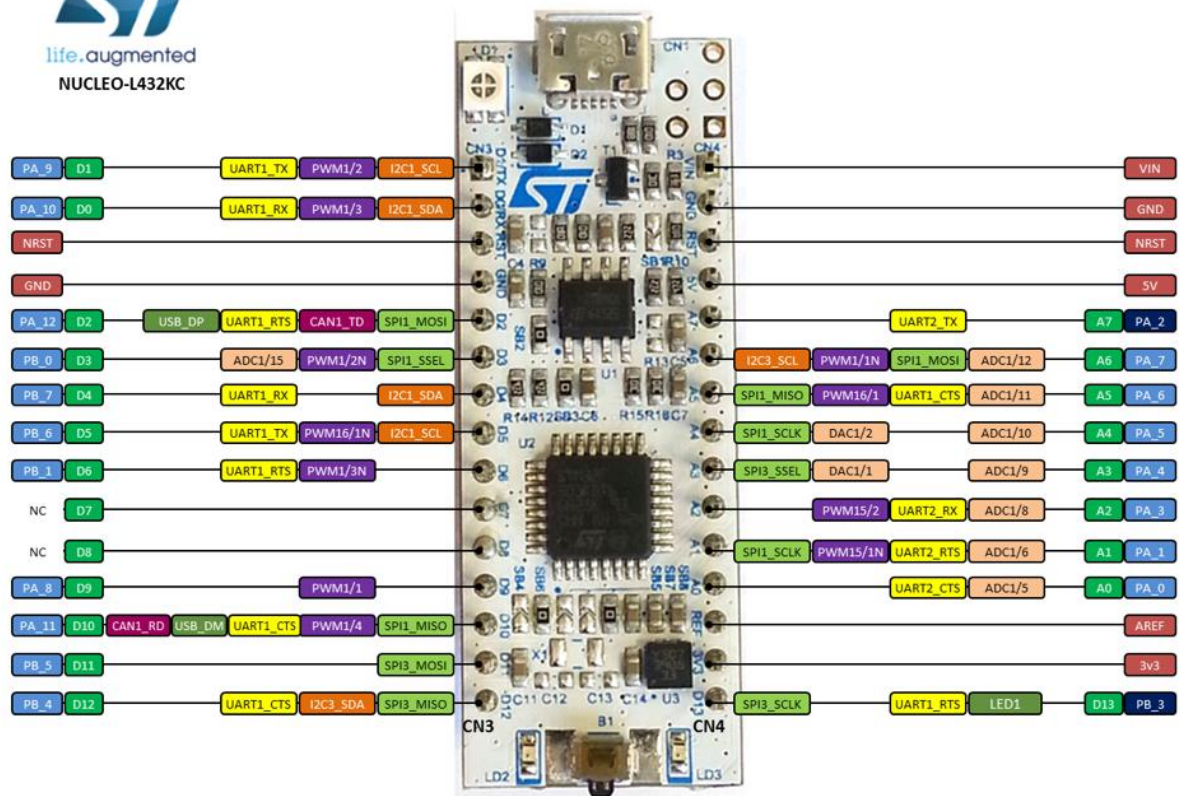
8. This assignment is actually wrong!
9. Right click on the pin, and select Signal unpinning



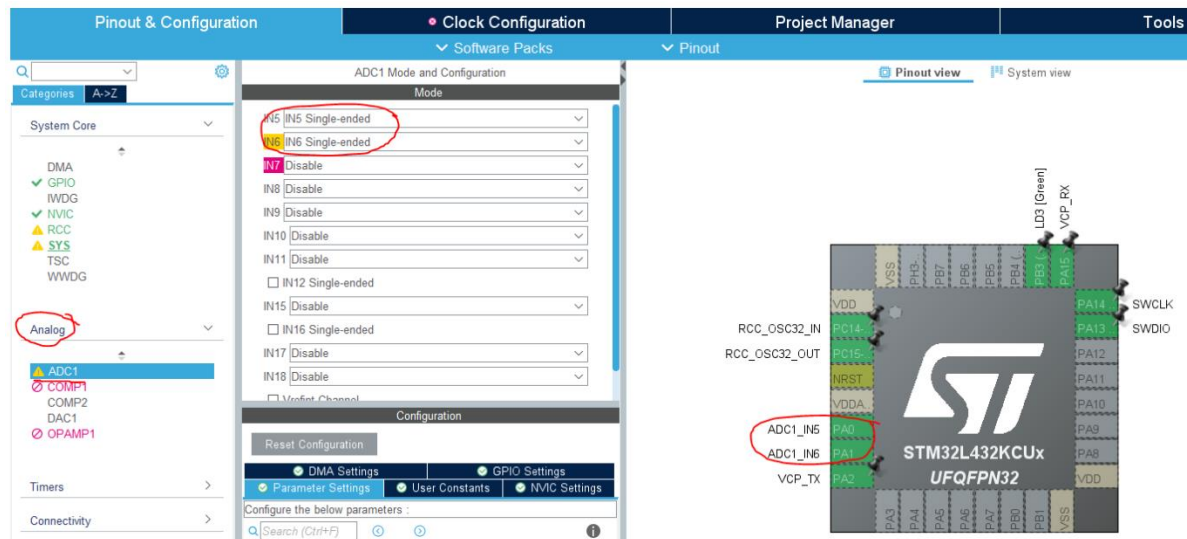
10. Click on System Core, RCC, then uncheck High Speed Clock HSE



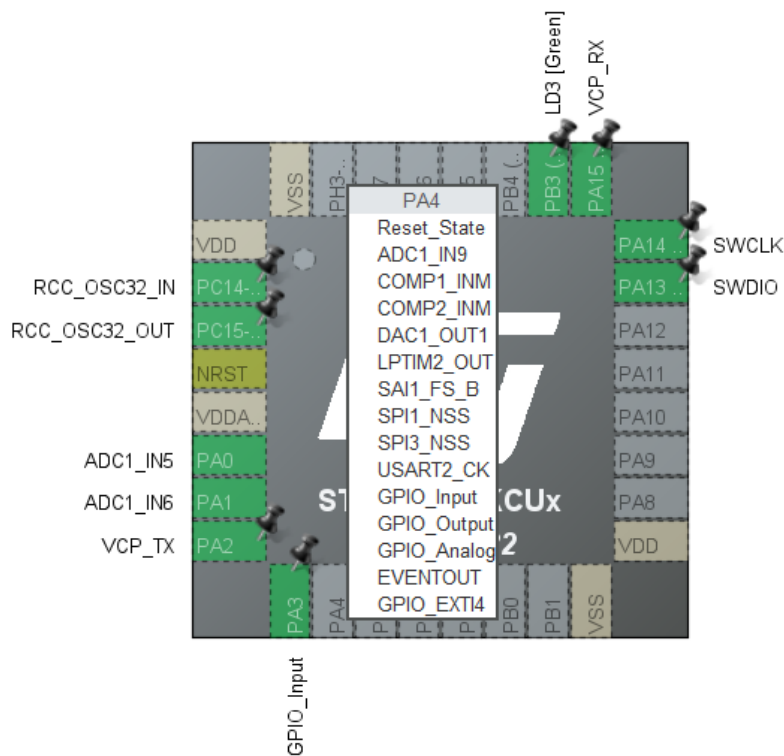
11. Configure your analogue pins. You must compare the pins on the Nucleo pinout with the port names:



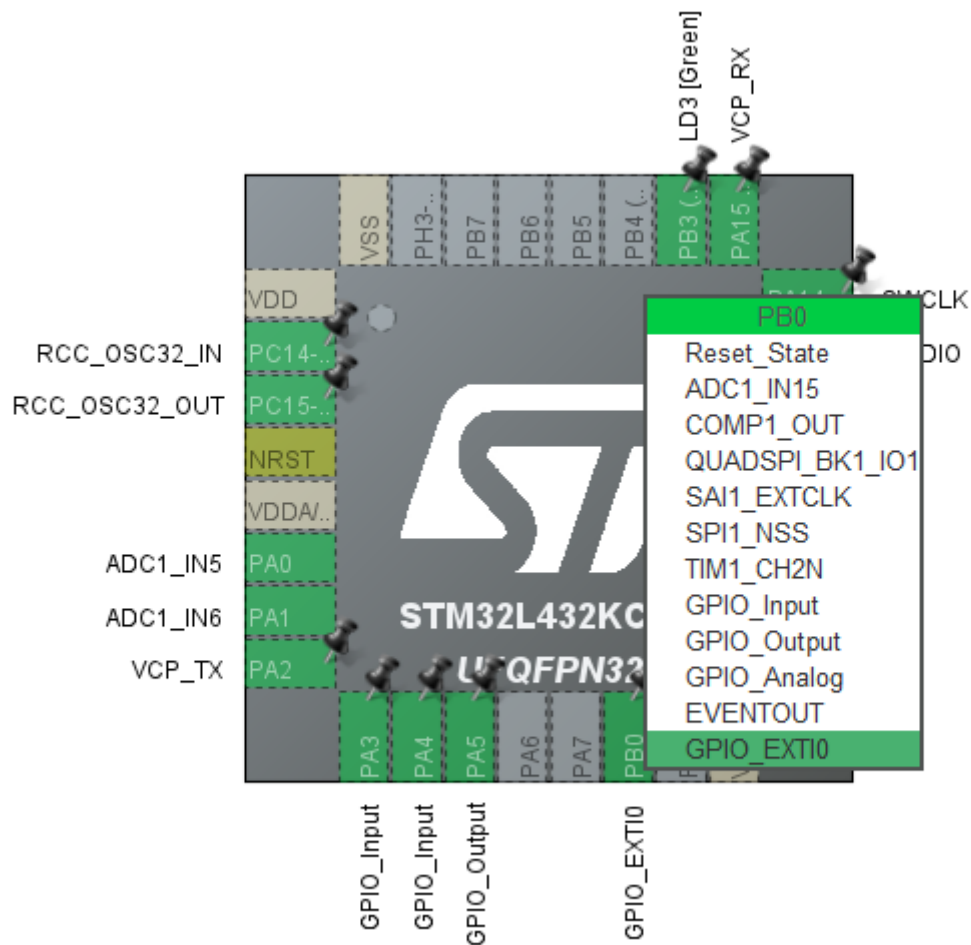
12. I have configured IN5 and IN6 as my analogue inputs. You have to check which pins you used for your PCB!



13. Click on all the pins that are GPIO inputs and outputs to configure them as such



14. Configure pins as GPIO_EXTIn if you want them to generate interrupts



- Check here and configure your outputs correctly as Low or High so that everything starts up in idle mode (Neither charging or discharging). Also check your interrupt and choose Rising or Falling edge

Pinout & Configuration

Categories: A-Z

System Core

DMA

☒ GPIO

IWDG

☒ NVIC

☒ RCC

☒ SYS

TSC

WWDG

Analog

ADC1

COMP1

COMP2

DAC1

OPAMP1

Timers

Connectivity

Multimedia

Security

Computing

Middleware and Software Packs

GPIO Mode and Configuration

Mode

Configuration

Group By Peripherals

☒ GPIO ☒ ADC ☒ RCC ☒ SYS ☒ USART ☒ NVIC

Search Signals

Search (Ctrl+F)

☐ Show only Modified Pins

Pin Na...	Signal on ...	GPIO out...	GPIO mode	GPIO Pull...	Maximum ...	Fast Mode	User Label	Modified
PA4	n/a	n/a	Input mode	No pull-up...	n/a	n/a		☐
PA5	n/a	Low	Output Pu...	No pull-up...	Low	n/a		☐
PB0	n/a	n/a	External I...	No pull-up...	n/a	n/a		☐
PB3 (JTD...	n/a	Low	Output Pu...	No pull-up...	Low	n/a	LD3 [Green]	☒

PB0 Configuration :

GPIO mode: External Interrupt Mode with Rising edge trigger detection

GPIO Pull-up/Pull-down: No pull-up and no pull-down

- You can use Pinout->Pins/Signals Options to name your pins via the User Label:

Pinout & Configuration

Categories: A-Z

System Core

DMA

☒ GPIO

IWDG

☒ NVIC

☒ RCC

☒ SYS

TSC

WWDG

Analog

ADC1

Pinout

Undo Mode and pinout (Ctrl-Z)

Redo Mode and pinout (Ctrl-Y)

☐ Keep Current Signals Placement (Ctrl-K)

☒ Show User Label

Disable All Modes (Ctrl-O)

Restore Pinout (Ctrl-N)

Clear Pinouts (Ctrl-P)

Clear Single Mapped Signals (Ctrl-M)

Pins/Signals Options... (Ctrl-O)

List Pinout Configuration settings for pins/signals

Export pinout with Alt. Functions (Ctrl-U)

Export pinout without Alt. Functions (Ctrl-U)

Reset used GPIOs (Alt-G)

Set unused GPIOs (Ctrl-G)

Pinout View Colors

Layout reset

MX Pins/Signals Options...			
	Pin Name	Signal Name	User Label
✖	PC14-OSC32_IN (PC14)	RCC_OSC32_IN	
✖	PC15-OSC32_OUT (PC15)	RCC_OSC32_OUT	
	PA0	ADC1_IN5	Vbatt
	PA1	ADC1_IN6	Vtemp
✖	PA2	USART2_TX	VCP_TX
✖	PA3	GPIO_Input	VbattHigh
✖	PA4	GPIO_Input	VbattLow
✖	PA5	GPIO_Output	ChargeON
✖	PB0	GPIO_EXTI0	FullChargeSwitch
✖	PA13 (JTMS-SWDIO)	SYS_JTMS-SWDIO	SWDIO
✖	PA14 (JTCK-SWCLK)	SYS_JTCK-SWCLK	SWCLK
✖	PA15 (JTDI)	USART2_RX	VCP_RX
✖	PB3 (JTDO-TRACESWO)	GPIO_Output	LD3 [Green]

17. When you have defined all you pins click on Clock Configuration and run the clock issues solver

The screenshot shows the STM32CubeMX interface. The 'Pinout & Configuration' window has the 'Clock Configuration' tab selected. A 'Resolve Clock Issues' button is visible. A dialog box titled 'Clock configuration' is open, asking 'Do you want to run automatic clock issues solver?'. The 'Yes' button is circled in red.

18. If it is still not OK, click the Resolve Clock Issues button

19. Save and Generate Code

20. If you put something like the following code in the main.c file

```
/* USER CODE BEGIN WHILE */

while (1)

{

    char msg[100];

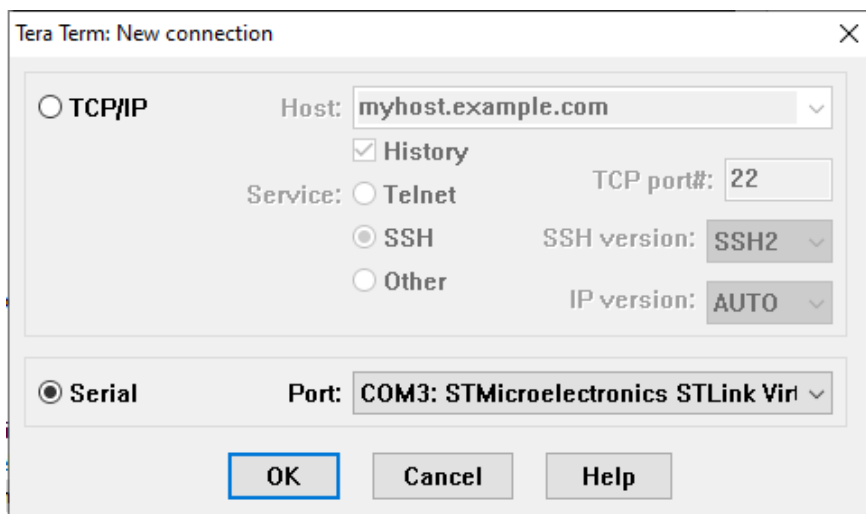
    sprintf(msg,"Hello World!\r\n");

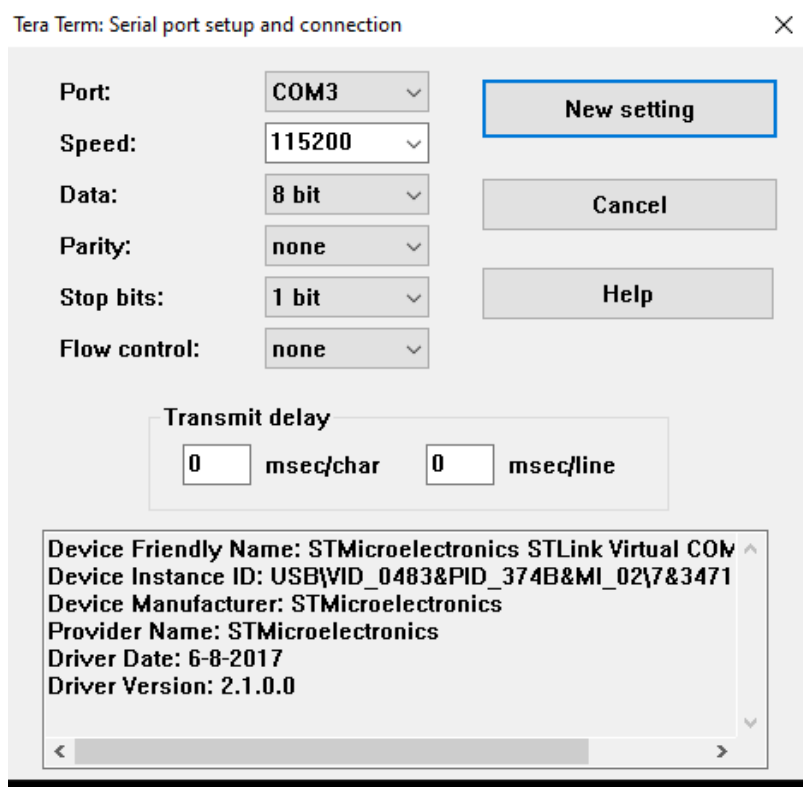
    HAL_GPIO_TogglePin(GPIOB, GPIO_PIN_3); // Toggle LD3

    HAL_UART_Transmit(&huart2, (uint8_t*)msg, strlen(msg), HAL_MAX_DELAY);

    HAL_Delay(1000); // Send every 1 second
```

Set up a terminal window for a serial com port. You should have the green LED blinking once per second, and Hello Wold! Sent to the terminal also every second.





19. Now we can test our two ADC channels:

```
/* USER CODE BEGIN WHILE */
```

```
while (1)
```

```
{
```

```
    char msg[100];
```

```
    uint32_t adc_value_CH5;
```

```
    uint32_t adc_value_CH6;
```

```
    ADC_ChannelConfTypeDef sConfig = {0};
```

```
    sConfig.Channel = ADC_CHANNEL_5; //First we will use channel 5
```

```
    sConfig.Rank = ADC_REGULAR_RANK_1; // Always rank 1 for single conversions
```

```
    sConfig.SamplingTime = ADC_SAMPLETIME_24CYCLES_5; // Example sampling time
```

```
    sConfig.SingleDiff = ADC_SINGLE_ENDED;
```

```
    sConfig.OffsetNumber = ADC_OFFSET_NONE;
```

```
    sConfig.Offset = 0;
```

```
    if (HAL_ADC_ConfigChannel(&hadc1, &sConfig) != HAL_OK)
```

```
    {
```

```
//Error_Handler();
}
```

```
HAL_ADC_Start(&hadc1); // Start ADC
HAL_ADC_PollForConversion(&hadc1, 100); // Wait for conversion
adc_value_CH5 = HAL_ADC_GetValue(&hadc1); // Read the result
HAL_ADC_Stop(&hadc1); // Stop ADC
```

```
sConfig.Channel = ADC_CHANNEL_6; //Then we will use channel 5
if (HAL_ADC_ConfigChannel(&hadc1, &sConfig) != HAL_OK)
{
//Error_Handler();
}
```

```
HAL_ADC_Start(&hadc1); // Start ADC
HAL_ADC_PollForConversion(&hadc1, 100); // Wait for conversion
adc_value_CH6 = HAL_ADC_GetValue(&hadc1); // Read the result
HAL_ADC_Stop(&hadc1); // Stop ADC
```

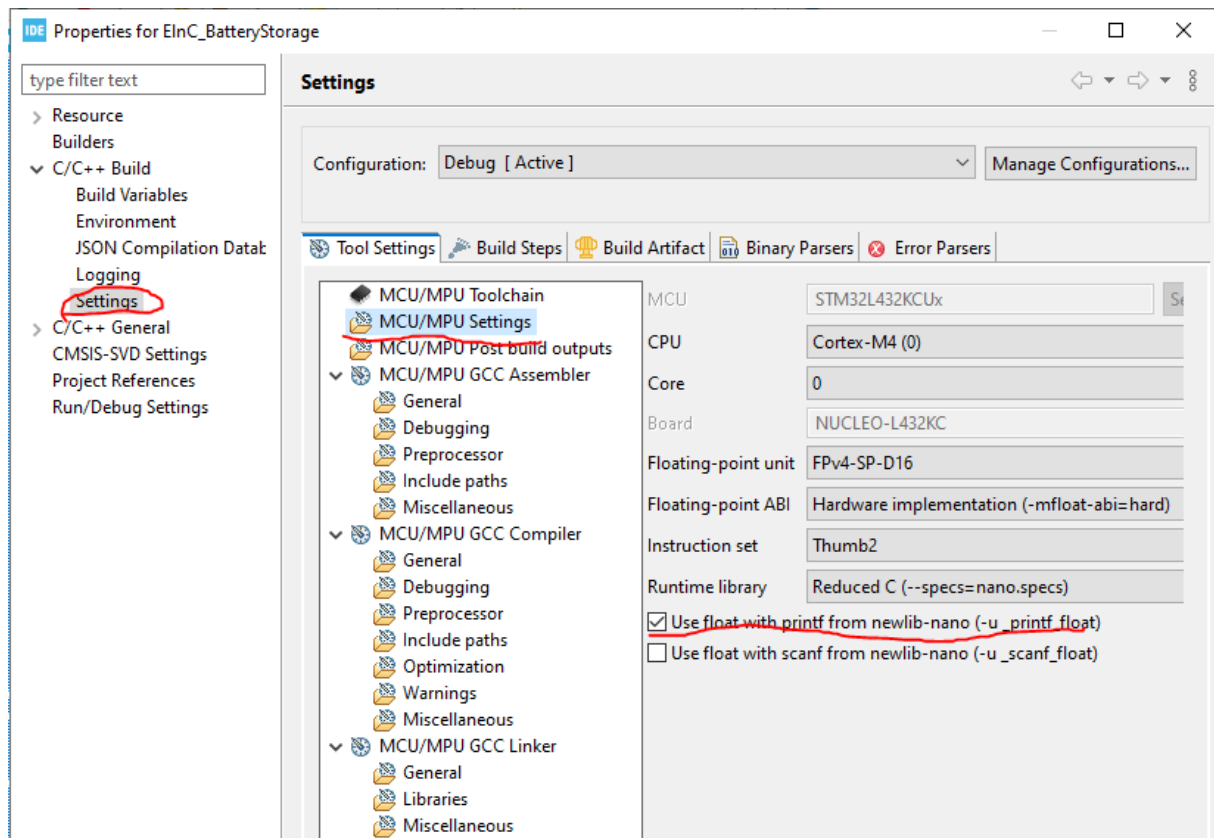
```
sprintf(msg, "ADC CH5 value is %ld\r\n", adc_value_CH5);
HAL_UART_Transmit(&huart2, (uint8_t*)msg, strlen(msg), HAL_MAX_DELAY);
```

```
sprintf(msg, "ADC CH6 value is %ld\r\n", adc_value_CH6);
HAL_UART_Transmit(&huart2, (uint8_t*)msg, strlen(msg), HAL_MAX_DELAY);
```

20. If you want to print floating point values, you must enable it. E.g.

```
sprintf(msg, "ADC CH5 value is %f\r\n", adc_value_CH5*Vref/4096.0);
```

Go to Project->Properties and check the following:



21. If everything works you can now start writing your code more correctly and better structured!